Definition

Let $\mathbf{G}$ be a set and $s, t : \mathbf{G} \to \mathbb{N}$ functions which specify the source and target of elements of $\mathbf{G}$. The set $E(\mathbf{G})$ of $\mathbf{G}$-*generated prop expressions* (or just *expressions*) is defined inductively as follows.

- ▹ The empty morphism $\mathrm{id}_0 : 0 \to 0$, the identity morphism $\mathrm{id}_1 : 1 \to 1$, and the symmetry $\sigma : 2 \to 2$ are expressions.
- ▹ The generators $g \in \mathbf{G}$ are expressions $g : s(g) \to t(g)$.
- ▹ If $f : a \to b$ and $g : x \to y$ are expressions, then $f + g : a + x \to b + y$ is an expression.
- ▹ If $f : a \to b$ and $g : y \to z$ are expressions, then $f \fatsemi g : a \to c$ is an expression.

The functions s and t are extended from $\mathbf{G}$ to $E(\mathbf{G})$ in the obvious way. If $f : m \to n$ is an expression, the pair $\langle m, n \rangle$ is called its *arity*.